COMPUTE ALLOCATION REPORT

NSF ACCESS

Explore · Discover · Accelerate · Maximize tiers on TACC, PSC, SDSC, NCSA, and partner systems

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Program Overview

The *Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support* (ACCESS) is the National Science Foundation's flagship cyberinfrastructure allocation program, launched in September 2022 as the operational successor to XSEDE. ACCESS coordinates open-science access to a federated portfolio of NSF-funded computing, storage, visualization, and cloud resources hosted at university and national laboratory partners. The program is administered through five service tracks (Allocations, User Support, Operations, Metrics, and Community Engagement & Performance) led by a consortium that includes the University of Illinois (NCSA), the Pittsburgh Supercomputing Center (PSC), the San Diego Supercomputer Center (SDSC), the Texas Advanced Computing Center (TACC), and others.

The federation includes some of the largest open-science systems in the United States. Representative member systems are TACC's *Stampede3*, *Vista* (Grace-Hopper GH200), and the NSF Leadership-Class *Frontera*; PSC's *Bridges-2* and *Neocortex* (Cerebras wafer-scale); SDSC's *Expanse* and *Voyager* (Habana Gaudi); NCSA's *Delta* and the AI-focused *DeltaAI*; Purdue's *Anvil*; and Indiana University's cloud-style *Jetstream2*. Allocations are denominated in a unified currency called *ACCESS Credits*, which users exchange at fixed per-resource rates for service units on individual systems. One ACCESS Credit is calibrated to roughly one CPU core-hour or one gigabyte-month of storage, with GPU-hours costing many credits depending on the device.

Hardware Specifications

The table below summarizes per-GPU characteristics for the ACCESS systems most relevant to deep-learning workloads. Dense FP16/BF16 tensor TFLOPS are reported with sparsity disabled; bandwidth and TDP are vendor-published values. *Neocortex* is included for completeness even though its Cerebras CS-2 wafer-scale engine is not GPU-comparable.

System	GPU	GPUs/ node	VRAM (GB)	FP16 TFLOPS	BW (TB/ s)	TDP (W)
NCSA Delta	A100 SXM4	4	40 / 80	312	1.55 / 2.0	400
NCSA Delta	A40 PCIe	4	48	149.7	0.696	300
NCSA DeltaAI	GH200 (H100)	4 (per node)	96 (HBM3)	756	4.0	700

System	GPU	GPUs/ node	VRAM (GB)	FP16 TFLOPS	BW (TB/ s)	TDP (W)
TACC Vista	GH200 (H100)	1 (super-chip)	96 (HBM3)	756	4.0	700
TACC Vista	H100 PCIe	2	80 (HBM3)	756	3.35	350
SDSC Expanse	V100 SXM2	4	32	125	0.9	300
Purdue Anvil	A100 SXM4	4	40	312	1.55	400
PSC Bridges-2	V100 SXM2	8	16 / 32	125	0.9	300
IU Jetstream2	A100	4	40	312	1.55	400
PSC Neocortex	Cerebras WSE-2	n/a	40 (SRAM)	*n/a, wafer-scale*	20 PB/s	23 kW

The H200 SXM reference (989 TFLOPS dense FP16, 141 GB HBM3e, 4.8 TB/s) used throughout this report exceeds every device above on FP16 throughput and on memory capacity, and exceeds all but the GH200 on bandwidth. Per-GPU FP16/BF16 ratios versus the H200 are approximately 0.32 (A100), 0.13 (V100), 0.15 (A40), and 0.76 (GH200/H100).

Allocation Tiers and Credit Conversion

ACCESS offers four project tiers. *Explore* is for graduate-student projects, course work, and pilot studies, with a cap of 400 000 ACCESS Credits and a streamlined eligibility-only review on a rolling basis. *Discover* supports research projects with modest needs and is capped at 1 500 000 credits, also reviewed on a rolling streamlined basis. *Accelerate* funds mid-scale collaborative work up to 3 000 000 credits and is reviewed by panels approximately every quarter. *Maximize* has no fixed credit cap and is awarded directly in resource units following a semiannual merit-review panel.

Tier	Credit cap	Duration	Review	Submission cadence
Explore	400 000	12 mo or grant	Eligibility	Rolling, hours-days
Discover	1 500 000	12 mo or grant	Streamlined	Rolling, days-weeks
Accelerate	3 000 000	12 mo or grant	Panel merit	Quarterly
Maximize	Variable, in-units	12 months	Panel merit, 10-page	Semiannual

Rolling submission for the lower tiers means that proposals can be submitted at any time and reviewed in the order received, rather than waiting for a fixed deadline window.

Worked credit-to-H200-hour conversion. On Delta, the published exchange rate places one A100 GPU-hour at one Delta GPU service unit, and the ACCESS Credit Calculator currently issues approximately 100 credits per Delta A100 GPU-hour. Spending *200 000* credits therefore buys roughly 2 000 A100 GPU-hours on Delta. The A100 delivers 312 dense FP16 TFLOPS, while the H200 delivers 989. The FP16-equivalent throughput is therefore

$$\text{H200-hours} \approx 2000 \times \frac{312}{989} \approx 631.$$

For Anvil GPU the calculator returns roughly 6 000 A100 GPU-hours per 400 000 credits (about 66.7 credits per GPU-hour), which scales to approximately 1 893 H200-hours of FP16 work. On DeltaAI the per-GH200-hour rate is roughly 250 credits, so 200 000 credits yield about 800 GH200-hours, or approximately 612 H200-FP16-equivalent hours.

Approval Timeline and Application Components

Explore reviews complete in hours to a few business days because the review is purely an eligibility check. Discover requests, which require a one-page abstract and a one-page resource justification, typically clear in a few business days to two weeks. Accelerate proposals require a three-page project description plus a CV and a prior-usage report and are panel-reviewed roughly every quarter, so wall-clock time from submission to award is typically four to six weeks. Maximize is the most demanding tier, with a ten-page project description, detailed scaling and benchmarking evidence, a prior-usage report, and CVs for all senior personnel; the semiannual panel cadence yields awards approximately three months after the submission window closes. All tiers require a brief renewal or progress narrative tied to the prior allocation.

Eligibility

Eligible principal investigators must be researchers, faculty, postdoctoral scholars, or staff at U.S.-based accredited degree-granting institutions, federally funded research and development centers, or non-profit research organizations. Graduate students may not serve as PI but can request and manage allocations under a faculty sponsor or advisor. All ACCESS-funded work must be *open science*, meaning results must be intended for unrestricted publication; export-controlled, ITAR, classified, or HIPAA-protected work is not permitted on standard ACCESS resources.

Things I Should Know

The most useful first-time trick is that any new account is automatically eligible for a small *startup* Explore allocation that can be requested with an abstract of just a few sentences, providing immediate access while a larger Discover or Accelerate request is being prepared. ACCESS storage is requested separately from compute, but a single project can hold one compute allocation per system and a co-located storage allocation, so users should request storage on the same site as their primary compute to avoid cross-site data movement. The *ACCESS Campus Champions* program embeds local cyberinfrastructure liaisons at most U.S. research universities; champions can sponsor startup allocations and shepherd new users through the proposal process.

Allocations can be *renewed* with a short progress narrative or *supplemented* mid-cycle if a project exhausts its credits early and can demonstrate productive use. Most major systems expose web gateways through

Open OnDemand and JupyterHub, which lower the barrier to interactive ML development. Export-control limits prohibit storing ITAR or EAR-restricted data on shared file systems. Default scratch retention is short (commonly 30 days untouched), and project file systems are purged at the end of an allocation unless data is migrated to long-term storage or to the user's home institution.

Strategic Fit for AI/ML Graduate Work

For AI/ML thesis work the practical question is how quickly a student can reach H100/H200-class throughput. *DeltaAI* is currently the strongest single answer in ACCESS: its quad-Grace-Hopper nodes provide H100-class tensor performance with 96 GB of HBM3 per GPU, NVLink-C2C to a 480 GB Grace CPU memory pool, and an exchange rate that makes a 400 000-credit Explore allocation translate to roughly 1 600 GH200 GPU-hours. *Vista* at TACC offers comparable per-GPU performance and is the second-fastest path. *Anvil* and *Delta* A100 partitions remain the workhorses for sub-100B-parameter training, fine-tuning, and reinforcement learning. *Expanse* and *Bridges-2* V100 partitions are best reserved for inference, smaller models, or capacity sweeps where 32 GB of VRAM suffices.

A successful ML proposal for ACCESS reviewers should foreground three things. First, a concrete scaling argument tying model size, batch size, and step time to the requested GPU-hours. Second, a clear training-versus-inference budget, since reviewers are sensitive to "I will use it for everything" requests. Third, a data-provenance and openness statement that confirms the work is publishable and that any datasets are not export-controlled. Compared to *NAIRR Pilot*, which currently emphasizes industry-donated H100 capacity, model API credits, and curated AI-ready datasets with a single coordinated review, ACCESS offers larger absolute GPU-hour budgets, longer time horizons, and more flexible compute-plus-storage bundling, at the cost of a heavier proposal burden at the Accelerate and Maximize tiers.

Summary Ranking

Tier	Approval	Typical size	H200-hr equiv.	Difficulty	Best use
Explore	Hours-days	400 k credits	1 200-1 600	Trivial	Pilots, coursework, thesis prelims
Discover	Days-weeks	1.5 M credits	4 700-6 000	Low	Single-PI ML research project
Accelerate	4-6 weeks	3 M credits	9 500-12 000	Moderate	Multi-student lab, mid-scale training
Maximize	3 months	Variable, in-units	10 000+	High	Large-scale pretraining, leadership runs

H200-hour equivalence is computed at dense FP16/BF16 tensor throughput (989 TFLOPS, 141 GB HBM3e, 4.8 TB/s) per H200 SXM GPU. Conversions below use the per-hour FP16 TFLOP ratio; memory-bound workloads scale differently and the bandwidth column is provided so readers can re-derive their own equivalence.

Sources

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